CAPSTONE PROJECT PRESENTATION

Free Sewaa

Community Donation & Resource-Sharing Platform

"Connecting Givers with Those in Need"

Introduction to Free Sewaa

A community-driven web platform designed to connect **givers** with those in **need**, fostering a culture of generosity and sustainability.

Promote Sustainability

Encouraging the reuse of usable items to reduce local household waste and environmental impact.

Community Support

Assisting students and low-income individuals by providing free access to essential resources.

Bridge the Gap

Creating a seamless digital link between generous donors and community members in need.

Community-First Approach

A dedicated platform focused purely on free item sharing rather than commercial transactions.

Problem Statement

The gap between abundance and necessity

The Central Issue:

Usable items are discarded while vulnerable community members struggle to afford basic necessities due to the lack of a centralized, non-commercial sharing platform.

Environmental Waste
Perfectly usable household items end up in landfills unnecessarily because donation is "too difficult."

Economic Barrier
Students and low-income families often cannot afford essential goods like textbooks or furniture.

Information Gap
No centralized local platform exists to find free resources or notify donors of nearby needs.

Commercial Noise
Existing platforms (Marketplace) prioritize commerce, burying free listings under paid advertisements.

"A community's surplus should be its neighbor's solution."

WASTE REDUCTION SOCIAL EQUITY

Proposed Solution

A Centralized Digital Hub for Community Generosity

Post Donations

Easily list usable items with descriptions and photos to reach local neighbors.

Browse & Search

Discover available resources nearby using smart categories and filters.

Request Items

Empower receivers to actively seek specific essentials they currently need.

Direct Coordination

Secure messaging system to arrange safe pickups and drop-offs.

- 100% Free Platform
- No-Commerce Focus
- Community Driven

Project Objectives

Defining the mission and goals of the Free Sewaa platform.

Functional Platform

Develop a robust web-based donation platform for community resource sharing.

Core Interactions

Enable users to seamlessly post, browse, and request donated items for free.

Direct Communication

Facilitate efficient dialogue between donors and receivers for item collection.

Waste Reduction

Promote sustainability and reuse to decrease local household waste levels.

Community Support

Provide essential goods to students and low-income families at no cost.

Intuitive Interface

Deliver a clean, accessible, and user-friendly experience for all skill levels.

Target Users

Identifying our primary community stakeholders

Students

- Access to textbooks and study aids
- Affordable furniture and kitchenware
- Support for campus living needs

Low-Income Individuals

- Seeking free essential goods
- Reducing financial burden
- Access to household necessities

Responsible Donors

- Wanting to give away usable items
- Promoting reuse over waste
- Supporting local social causes

Community Members

- Participation in sustainable sharing
- Strengthening local connections
- Reducing neighborhood waste

Key Platform Features

Secure Authentication
Robust registration and login system for secure account management.

Item Posting
Simplified form to list donations with photos and descriptions.

Browse & Search
Searchable and filterable grid view of all available community items.

Request System
Enables users to actively request specific essential goods they need.

Direct Communication
Integrated messaging system to arrange pickups and exchange details.

Intuitive Interface
Clean, responsive, and accessible design for all user demographics.

System Architecture

A Robust Three-Tier Web Framework

Presentation Layer

React.js

- Responsive UI components
- Client-side routing
- Dynamic state management

Application Layer

Node.js / Express.js

- RESTful API endpoints
- User authentication (JWT)
- Business logic processing

Data Layer

MongoDB / MySQL

- User profile storage
- Item listing management
- Messaging data persistence

User Interaction > Frontend (React) > REST API (Node/Express) > Database Storage

Technology Stack

FRONTEND

- React.js
- HTML5 / CSS3
- JavaScript (ES6+)

BACKEND & DATA

- Node.js / Express.js
- MongoDB / MySQL
- REST API Architecture

DEV TOOLS

- GitHub(Version Control)
- Postman (API Testing)
- Figma (UI/UX Design)

DEPLOYMENT

- Vercel(Frontend)
- Render(Backend)
- Continuous Integration

UI & Interface Design

A clean, professional blue-and-white themed visual experience.

Homepage & Nav

- Centralized search bar for quick item discovery.
- Featured donation highlights on landing.
- Intuitive top-level navigation menu.

Item Listing Grid

- Responsive grid view with large photo previews.
- Category tags (Furniture, Books, etc.) for filtering.
- Clear availability status indicators.

User Dashboard

- "My Listings" panel for managing donations.
- Integrated messaging for donor communication.
- Request history and status tracking.

DESIGN PRINCIPLES

- Clean Typography
- High Contrast
- Intuitive Flow

Project Workflow

The structured development lifecycle of Free Sewaa

PHASE 01

Requirements
Scope definition & needs analysis

PHASE 02

UI/UX Design
Figma wireframes & mockups

PHASE 03

DB Schema
Data structure & relations

PHASE 04

Frontend
React component building

PHASE 05

Backend API
Node/Express logic

PHASE 06

Integration
System testing & debugging

PHASE 07

Deployment
Vercel & Render hosting

PHASE 08

User Testing
Feedback & final polish

TECH ECOSYSTEM: GitHub Figma Postman

Iterative Development Methodology

Current Progress

Development Status & Implementation Roadmap

Completed

- Project Planning & Scope
- UI/UX Wireframing
- Database Schema Design
- User Auth (Login/Reg)
- Item Posting Form

In Progress

- Browse & Search Logic
- Item Request System
- In-app Messaging Feature

Overall Implementation 75%

Upcoming

- Full Integration Testing
- Production Deployment
- User Acceptance Testing

Challenges & Mitigations

User Adoption

Encouraging initial community engagement and platform usage.

- Mitigation: Targeted outreach to student groups & social media.

Item Verification

Ensuring donated items are in usable and safe condition.

- Mitigation: Rating system and clear photo requirements.

Real-Time Chat

Implementing reliable communication between donors/receivers.

- Mitigation: Using Socket.io for efficient message delivery.

Platform Security

Protecting user data and preventing platform misuse.

- Mitigation: JWT authentication & data encryption protocols.

Future Improvements

A roadmap of planned enhancements to expand Free Sewaa's reach, intelligence, and accessibility.

Geolocation Integration

Display nearby donations based on the user's real-time location for faster pickups.

Push Notifications

Alert users instantly when new items matching their interests become available.

Mobile Application

Expand platform access with native iOS and Android apps for on-the-go usage.

AI-Based Recommendations

Deliver personalized item suggestions to users based on their browsing and request history.

Multi-Language Support

Make the platform inclusive and accessible to diverse communities across languages.

Donor Rating System

Build community trust through verified user reviews and transparent donor ratings.

Conclusion

Social Responsibility

Addresses real community needs by providing a free, accessible platform for essential goods distribution.

Environmental Impact

Promotes sustainability and waste reduction by facilitating the reuse of household items.

Technical Execution

Demonstrates full-stack proficiency using modern, scalable web technologies (MERN/MySQL).

Thank You!

Questions & Discussion

"Connecting Givers with Those in Need"